**Abbreviations**

| | |
|---|---|
| cao | correct answer only |
| dep | dependent |
| FT | follow through after error |
| isw | ignore subsequent working |
| oe | or equivalent |
| SC | Special Case |
| nfww | not from wrong working |
| soi | seen or implied |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | 607 021 | **1** | |
| 2 | Correct shape drawn | **1** | |
| 3(a) | Correct reason | **1** | |
| 3(b) | 21.5 oe | **1** | |
| 4 | 02 11 | **1** | |
| 5 | −31 | **1** | |
| 6 | $11h - 2w$ final answer | **2** | **M1** for $11h + kw$ or $kh - 2w$ |
| 7(a) | Kite or isosceles trapezium | **1** | |
| 7(b) | Correct shape drawn | **2** | **M1** for 3 sides correctly reflected or 4 correct vertices |
| 8(a) | 72<br>Corresponding angles | **2** | **B1** for each |
| 8(b) | 65<br>Angles [at a point] sum [to] 360 oe | **2** | **B1** for each |
| 9(a) | 32 | **1** | |
| 9(b) | 36 | **1** | |
| 9(c) | 37 | **1** | |
| 10(a) | Circle with 3.8 cm radius drawn | **2** | **M1** for 11.4 ÷ 1.5 or 5.7 ÷ 1.5 |
| 10(b) | 4.8 | **2** | **M1** for $\frac{9.6\times50000}{100\times1000}$ or answer figs 48 |
| 11 | 116 | **2** | **M1** for angle $ACB = 32$ soi |
| 12 | 90 | **2** | **M1** for $\frac{42}{7}\left[\times(7+3+5)\right]$ soi by 6 |
| 13(a) | 0.07 | **1** | |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 13(b) | 14 | **1** | **FT** *their* **(a)** |
| 14(a) | $p^6$ | **1** | |
| 14(b) | $m^{10}$ | **1** | |
| 14(c) | $k^{15}$ | **1** | |
| 15 | Rotation<br>clockwise 90°<br>[centre] (0, 0) | **3** | **B1** for each |
| 16 | Correct common denominator | **M1** | |
| | Correct method | **M1** | e.g. $1\frac{3-8}{12}$ or $\frac{12+3-8}{12}$ or<br>$\frac{((3\times4)+1)\times3-((2\times3)+2)\times4}{12}$ or $\frac{39-32}{12}$ |
| | $\frac{7}{12}$ cao | **A1** | |
| 17 | 7.3 cao | **4** | **M3** for $\frac{150\times0.84}{3.45\times5}$<br>or **B3** for final answer figs 73[04...]<br><br>or **M2** for $\frac{150\times0.84}{3.45}$ or $\frac{3.45\times5}{0.84}$<br>or **M1** for $150\times0.84$ or $3.45\times5$ or $\frac{3.45}{0.84}$ |
| 18 | A 46.28 | **5** | **M1** for $5000\times1.035^6$<br>**A1** for 6146.27 to 6146.28<br>**M1** for $5000\times1.22$<br>**A1** for 6100 |
| 19 | $\frac{70\times30}{5^2}$<br>and 84 | **2** | **M1** for two of 70, 30, 5 used |
| 20 | 28, –34 | **4** | Trial and improvement<br><br>OR<br><br>**B1** for $x+y=-6$ oe<br>**B1** for $x-y=62$ oe<br>**B1** for 28 or –34 |